

PROJECT SUTRA

Evaluation 1 — Jury Defense Playbook

Smart Horizon 48-Hour International Hackathon
Defence & SpaceTech Track — SH-DST-05

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I. How Evaluation 1 Works

Jury psychology, scoring breakdown, and what happens in the room

The Room Setup

Eval 1 happens on **Day 1 at 5:00 PM**. The jury has already seen 3–4 other teams. They are tired, skeptical, and looking for reasons to filter. Your window is **15–20 minutes total**: ~5 minutes for your pitch, ~10–15 minutes for Q&A.

What the Jury Evaluates (100 Marks)

Criteria	Marks	What They Actually Look For
System Architecture & Design	25	Can the team explain their system from first principles? Do the subsystems connect logically?
Problem Statement Alignment	20	Does the solution actually solve SH-DST-05? Or is it a generic drone project with SAR labels pasted on?
Baseline Prototype Execution	25	Does the code actually run? Can they demo it? Are the tests real?
Technical Innovation & Math Rigor	15	Can they derive the math? Or did they copy-paste from a paper?
Workstation Discipline & Pitch	15	Git hygiene, branch structure, pitch clarity, timing.

Jury Psychology — What They're Really Thinking

1. **"Is this team for real?"** — They scan for rehearsed answers vs. genuine understanding.
2. **"Who's the real lead?"** — They watch who speaks first and who defers. If one person answers everything, they probe the others.
3. **"Where's the gap?"** — They pick the weakest-looking claim and drill until they find the wall.
4. **"Can they defend under pressure?"** — They intentionally misstate something to see if the team corrects them or agrees.
5. **"Is this deployable?"** — They think like operators, not academics. "Would I trust this with my life?"

The First 60 Seconds

The jury walks in. They don't ask questions immediately. They are:

- Looking at your screen setup (Gazebo running? Terminal open? GCS visible?)

- Reading your git branches (clean? organized? or messy?)
- Watching team body language (who's nervous? who's confident?)
- Scanning your pitch deck or DOCS.md

Critical: Do NOT rush to explain before being asked. Stand calmly. Let your setup speak. When they ask "Tell us about your project," that's your 60-second elevator pitch — not a 10-minute lecture.

II. Phase 1: Architecture Deep Dive

Questions 1–12 | Tests: 25-mark Architecture criterion | Difficulty: Medium to Hard

Q1: "Walk me through what happens from the moment a drone detects an obstacle to the moment the motor command changes."

Testing: End-to-end architectural understanding, latency awareness, data flow mastery

GROUNDING ANSWER

The pipeline has 6 stages, each with a measured latency:

1. **Sensor ingestion (0.5ms):** LiDAR PointCloud2 arrives on `/drone_id/points` at 10Hz. The `octomap_generator.py` node inserts voxels into a 32×32×16 grid at 0.10m resolution.
2. **Occupancy update (0.2ms):** `sutra_fsd_occupancy.py` applies temporal decay ($\lambda=0.92$) and marks occupied voxels with color-coded altitude.
3. **Trajectory planning (0.8ms):** `sutra_fsd_trajectory_planner.py` evaluates 35 candidate quintic polynomial ribbons, scoring each by obstacle proximity and jerk cost. Best ribbon selected.
4. **CBF safety filter (0.06ms):** `sutra_cbf_safety_shield.py` solves a QP projecting the acceleration command onto the safe half-space where $h(x) = \|p_i - p_j\|^2 - R^2 \geq 0$.
5. **ORCA avoidance (0.5ms):** `orca_avoidance.py` computes reciprocal velocity obstacles for all 5 peers and clamps the preferred velocity into the ORCA half-plane intersection.
6. **PX4 setpoint publish (0.1ms):** `px4_offboard_controller.py` publishes TrajectorySetpoint at 50Hz over MicroXRCE-DDS.

Total感知到执行延迟: ~2.2ms within the 20ms budget at 50Hz.

Follow-up 1: "What's the latency budget? Add it up for me."

Answer: "About 2.2ms total pipeline, well within our 20ms budget at 50Hz. The bottleneck is trajectory planning at 0.8ms, which is still under 1ms."

Follow-up 2: "Where exactly in that chain could a packet drop cause a collision?"

Answer: "Stage 6 — the MicroXRCE-DDS publish. If the setpoint doesn't reach PX4, the drone holds its last velocity for one cycle. At 3 m/s, that's 6cm of drift. The CBF safety filter in stage 5 already accounts for this by maintaining a 2.80m buffer, which gives us 930ms of margin before collision."

Tone: Methodical and calm. Walk through each stage like a surgeon explaining a procedure. Use your hands to count the stages. Don't rush — the jury wants to see you can recall this from memory, not from notes.

Q2: "Your ORCA solver — is it centralized or decentralized? What's the communication overhead?"

Testing: Distributed systems understanding, mesh awareness, scalability thinking

GROUNDING ANSWER

Fully decentralized. Each drone runs its own `Orca3DSolver` instance. There is no central coordinator. Here's how it works:

- Each drone subscribes to `/peer_id/odometry` for all 4 peers (see `orca_avoidance.py:490-505`).
- The solver computes avoidance velocities using only local peer positions and velocities — no consensus needed.
- **Communication overhead:** Each drone publishes its odometry at 10Hz (not 50Hz — avoidance doesn't need 50Hz peer updates). That's 5 drones × 10Hz × ~200 bytes = 10 KB/s total mesh traffic for swarm coordination.

The **SwarmRAFT** consensus layer (in `mesh_node.py`) operates independently for leader election and task allocation — it does NOT gate the ORCA loop. ORCA works even if the mesh is completely partitioned.

Follow-up 1: "If drone 3 can't reach drone 5 over mesh, does ORCA still work?"

Answer: "Yes. ORCA only needs the last-known position and velocity of each peer. If mesh drops for 2 seconds, we have 2 seconds of stale odometry. The ORCA time horizon $\tau = 1.5s$ means we only look 1.5 seconds ahead. As long as we received peer data within the last 1.5s, avoidance is valid. After that, we switch to conservative stop."

Follow-up 2: "What's the worst-case computation time for 5 agents? What about 50?"

Answer: "For 5 agents, each drone solves a linear program with 4 half-plane constraints — under 0.5ms on CPU. For 50 agents, it's 49 constraints per drone. Our `test_huge_swarm_coordination.py` tests up to 50 drones and the solver stays under 1.0ms. The complexity is $O(N)$ per drone, not $O(N^2)$."

Tone: Confident and precise. This is a strength question — ORCA is well-tested (120 tests pass). Lean into the data.

Q3: "You mention Control Barrier Functions. Write the barrier condition on the whiteboard. What does γ control physically?"

Testing: Math depth, first-principles understanding, ability to derive under pressure

GROUNDING ANSWER

Write on whiteboard:

$$h(x) = \|p_i - p_j\|^2 - R_{\text{safe}}^2 \geq 0$$

$$\dot{c}(x) + \gamma h(x) \geq 0$$

Where:

- **$h(x)$** is the safety barrier — it's positive when drones are far apart, negative when they're too close.
- **R_{safe}** = 2.80m (our hard minimum clearance).
- **γ (gamma)** controls how aggressively we push drones apart as they approach the barrier. Physically, it's the *reactance coefficient* — higher γ means stronger repulsion at greater distances.

What γ means in practice:

- γ too small (e.g., 0.1): Drones get very close before repulsion kicks in. Unsafe.
- γ too large (e.g., 10.0): Drones repel each other from far away, wasting energy and disrupting search patterns.
- We use **$\gamma = 2.0$** — tested in `test_sutra_fsd_autopilot.py` as the CBF hard barrier parameter.

Follow-up 1: "What happens if γ is too large? Too small?"

Answer: "Too large makes the swarm overly conservative — drones maintain 5m+ separation even when the mission requires tight formation. Too small allows approach below R_{safe} before the QP solver can project the command. We sweep γ from 0.5 to 5.0 in simulation and $\gamma=2.0$ gives the best trade-off between safety and mission performance."

Follow-up 2: "Is your CBF actually solving a QP at runtime? What solver?"

Answer: "Yes, at every 50Hz tick. We use a simple inequality constraint projection — since our barrier is quadratic and we have linear dynamics, the QP has a closed-form solution for 2-agent

pairs. For 5 agents, we solve 4 independent QPs sequentially. Total time: 0.06ms. No external QP solver library needed."

Tone: Teach like a professor. Write slowly on the whiteboard. Make eye contact after writing. The jury wants to see you OWN the math, not recite it.

Trap: If they ask "Is this from a paper?" — say "The CBF formulation comes from Ames et al. 2014, but our specific implementation with the closed-form QP for multi-agent reciprocal barriers is original work, implemented in `sutra_cbf_safety_shield.py`."

Q4: "Why Quintic Splines? Why not Cubic or Trapezoidal?"

Testing: Control theory depth, understanding of physical rotor dynamics

GROUNDING ANSWER

Three reasons, in order of importance:

1. **C² continuity:** Quintic splines give continuous position, velocity, AND acceleration. Cubic splines only give C¹ (continuous velocity), meaning acceleration jumps discontinuously. That step change in acceleration = infinite jerk = motor saturation and ESC thermal damage.
2. **Jerk bounding:** Our planner explicitly minimizes $\int \|j(t)\|^2 dt$. We enforce $\text{jerk} < 4.2 \text{ m/s}^3$. This is testable — `test_sutra_fsd_autopilot.py` verifies the quintic spline output has bounded jerk.
3. **Boundary constraints:** We need to specify p_0, v_0, a_0 AND p_T, v_T, a_T — 6 boundary conditions. A quintic polynomial has exactly 6 coefficients (a_0 through a_5). A cubic only has 4 — not enough to specify acceleration at both endpoints.

Follow-up: "What's the condition number of the boundary matrix? Is it well-conditioned?"

Answer: "The 6x6 matrix is a Vandermonde-like system. For $T = 1.0\text{s}$, the condition number is around 16 — well-conditioned. For very large T ($> 10\text{s}$), it becomes ill-conditioned, but our trajectory segments are always 0.5–2.0 seconds."

Tone: Start with the physical consequence ("infinite jerk destroys motors"), then connect to the math. The jury remembers stories, not equations.

Q5: "You claim 96.9% compression with Deep JSCC. Walk me through exactly what's transmitted over the air."

Testing: Communications depth, understanding of JSCC vs. traditional codecs

GROUNDING ANSWER

Here's the exact pipeline, stage by stage:

1. **Input:** A 320×256 thermal frame = 81,920 pixels × 1 byte = ~80 KB raw.
2. **Encoder:** `PerceptronJSCCEncoder` (in `perceptron_jsc.py`) is an MLP: 512 → 128 → 16 with BatchNorm and Tanh power normalization. It compresses to a 16-dimensional latent vector.
3. **Channel mapping:** The 16 floats are mapped to 16 complex symbols (I/Q pairs) for OFDM subcarrier transmission. Each symbol is normalized by the Tanh layer to fit within transmit power constraints.
4. **Output:** 16 × 8 bytes (float64) = 128 bytes, or 16 × 4 bytes (float32) = 64 bytes. From 80 KB to 64 bytes = **99.9% raw compression**. The "96.9%" figure in our docs accounts for the full frame including headers and metadata.

What the receiver gets: 16 complex symbols. The `PerceptronJSCCDecoder` (16 → 128 → 512 MLP) reconstructs a 512-dimensional feature vector. This is NOT a pixel-perfect image — it's a *semantic feature representation* that preserves human heat signatures for the operator.

Follow-up 1: "So you're sending 16KB of what? How do you map to RF?"

Answer: "16 complex symbols, each mapped to an OFDM subcarrier. We don't use a separate modulation scheme — the neural network learns the modulation implicitly. That's the key insight of JSCC: there's no separate QAM block. The encoder outputs symbols that are already optimized for the channel."

Follow-up 2: "Is it actually analog or digital with soft symbols?"

Answer: "It's digital in implementation (floating-point symbols over digital radio) but analog-like in behavior. The symbols are continuous-valued, not quantized to a finite constellation like QPSK or 16-QAM. This is why it degrades gracefully — there's no hard decision boundary to fail at."

Tone: Be precise about what IS and what IS NOT transmitted. If they sense you're hand-waving about the RF layer, they'll push harder. Admit: "The actual RF modulation is handled by the Wi-Fi mesh adapter — our neural network outputs baseband symbols."

Q6: "You claim 41.5 dB PSNR at -5 dB SNR. How did you measure that? Show me the test."

Testing: Benchmark integrity, Zero-Mock Policy compliance

GROUNDING ANSWER

This is a **critical honesty moment**. Here's the precise truth:

The 41.5 dB PSNR was measured using our `PerceptronSemanticCommsPipeline` in `perceptron_jscc.py`. The PSNR is computed as:

- The encoder processes a 512-dim feature vector (extracted from a ResNet18 backbone in training).
- Gaussian noise is added to simulate the channel at the specified SNR.
- The decoder reconstructs the feature vector.
- MSE is computed between original and reconstructed features using `nn.functional.mse_loss`.
- $\text{PSNR} = 10 \times \log_{10}(\text{MAX}^2 / \text{MSE})$.

Honest caveat: This is feature-space PSNR, not pixel-space PSNR. We have NOT run the full encoder-decoder on actual camera images at -5 dB in real-time. The training script (`train_deep_jscc.py`) and evaluation script (`evaluate_jscc_on_real_images.py`) exist and can be run, but the live inference on actual thermal frames at -5 dB SNR is not yet benchmarked end-to-end.

Follow-up: "If I connect your JSCC encoder to a real SDR right now, will it work?"

Answer: "Not yet. The neural network is trained and the inference pipeline runs on GPU, but we haven't integrated with a physical SDR. What we CAN show is the PSNR degradation curve across SNR values in simulation — `test_deep_jscc_neural_audit.py` tests compression ratio and PSNR across the full SNR spectrum."

Trap: If you claim the 41.5 dB is pixel-space PSNR measured on real images, and the jury asks you to show the reconstruction — you'll be caught. Be honest: "This is feature-space PSNR. The full pixel-space benchmark is on our roadmap."

Tone: Honest and precise. Hackathon jurors RESPECT teams that know the difference between "we measured X" and "we can show Y." Saying "we haven't tested that yet but here's what we HAVE tested" is stronger than bluffing.

Q7: "Your WGS84 raycaster — what happens when the drone is at 30m altitude and tilted 25 degrees?"

Testing: Geolocation math depth, understanding of 3D projection

GROUNDING ANSWER

The full formula is in `detector_node.py:160-192` (`to_gps` function):

$$p_{\text{world}} = p_{\text{UAV}} + s \cdot R_b^w \cdot K^{-1} [u, v, 1]^T$$

1. **Camera ray:** We take the 2D bounding box center (u, v) and multiply by K^{-1} (inverse intrinsic matrix) to get a normalized camera-frame ray.

2. **Rotation:** We apply the full body-to-world rotation matrix R_b^w using the drone's IMU quaternion. At 25° pitch, the rotation matrix tilts the ray forward significantly.
3. **Raycast:** We shoot the ray from the drone's GPS position and find the intersection with the terrain DEM $Z(X,Y)$ using parametric ray marching.
4. **WGS84 conversion:** The intersection point is converted to lat/lon using Earth radius $R = 6,378,137\text{m}$ with cosine latitude correction for longitude.

At 25° tilt: The ray hits the ground ~12m forward of the drone's nadir point. Without tilt compensation, a flat 2D projection would place the target at the nadir — error of ~12m. Our 3D raycast eliminates this error. Tested: `test_detector.py` includes tilt scenarios up to $\pm 45^\circ$ with mean error $< 0.04\text{m}$.

Follow-up: "What's the DEM resolution? Where did you get the elevation data?"

Answer: "For SITL, we use a flat DEM ($Z=0$). For real deployment, we'd use SRTM 30m DEM or LiDAR-generated local DEM. The raycaster is resolution-agnostic — it just needs $Z(X,Y)$ at any grid spacing."

Tone: Walk through the formula step by step. Point to the exact function in the code: "Let me show you — `detector_node.py`, line 160."

Q8: "Your Tri-Modal fusion — when all three sensors disagree, who wins?"

Testing: Sensor fusion logic, failure mode understanding

GROUNDNED ANSWER

The fusion logic is in `detector_node.py:769-886` (`_fusion_tick`):

1. **Visual detections seed the fusion** — they have the highest info content and best spatial resolution.
2. **Thermal cross-checks via IoU** — if a thermal blob overlaps a visual detection ($\text{IoU} > 0.15$), the thermal confidence boosts the score.
3. **Radar confirms proximity** — if radar clusters a target within 3m ground radius, it confirms the detection.
4. **Fallback:** If RGB confidence drops below 0.20 (darkness, smoke), thermal-only detections pass through with reduced weight.

Weighted scoring: $\text{score} = \text{conf} \times 0.50 + \text{thermal} \times 0.35 + \text{radar} \times 0.15$. Classification thresholds: $\text{SURVIVOR} \geq 0.60$, $\text{POSSIBLE_SURVIVOR} \geq 0.30$.

When they disagree: If thermal sees a heat signature but YOLO says "rock" ($\text{conf} < 0.20$), the thermal-only fallback triggers. If radar sees nothing but visual and thermal agree, the detection still passes with reduced confidence. The system is designed to be *recall-biased* — better to have a false positive than miss a survivor.

Tone: Be specific about the weights and thresholds. The jury wants to see you've thought about edge cases, not just happy-path fusion.

Q9: "You say 60 FPS WebGPU. What's actually rendering at 60 FPS? The map? The video? The HUD overlay?"

Testing: Frontend engineering depth, WebGPU vs Canvas2D understanding

GROUNDING ANSWER — HONEST VERSION

This is a **known gap** we need to address honestly.

What we claim: "WebGPU Telemetry HUD Locked 60 FPS" (Gate G6).

What's actually implemented: The GCS dashboard in `sutra_gcs/src/` currently uses **Canvas 2D API** (`canvas.getContext( '2d' )`) for the physics simulations and HUD overlays. The Mapbox GL JS dependency is installed but not yet imported — the "3D GIS" view is a CSS-based radar grid.

What we CAN demonstrate:

- The React 18 app builds successfully (`npm run build` passes).
- The Canvas 2D physics simulations (SORCA ring crossing, swarm comms) render at 60 FPS in the browser.
- The WebSocket client connects to the ROS 2 gateway and receives telemetry.

What's on the roadmap: Integrating Mapbox GL JS for 3D terrain and migrating the HUD to WebGPU compute shaders. This is the #1 priority before Eval 3.

Follow-up: "So your 60 FPS claim is Canvas 2D, not WebGPU?"

Answer: "Correct. The Canvas 2D rendering hits 60 FPS for the current demo load. WebGPU migration is planned for the 48-hour sprint. The architecture is designed for it — we already have ring buffer telemetry (`telemetryBuffer.ts`) that feeds binary `ArrayBuffers`, which are WebGPU-ready."

Trap: If you claim "yes, WebGPU is running" and the jury asks to see the GPU compute shader code — there is none. Be honest. "We're transparent about what's implemented vs. planned. The Canvas 2D demo is functional and we're upgrading to WebGPU during the hackathon."

Tone: Confident honesty. "Here's what we've built, here's what's in progress." Juries reward teams that know their gaps over teams that bluff.

Q10: "Why 50Hz setpoints? Why not 10Hz or 100Hz?"

Testing: PX4/offboard control understanding, real-time systems knowledge

GROUNDING ANSWER

- **10Hz is too slow:** At 4 m/s cruise speed, a drone travels 40cm between updates. ORCA needs relative velocity updates faster than the time-to-collision. At 3m separation and 4m/s closing speed, that's 750ms to collision — 10Hz gives only 7.5 updates, which is borderline.
- **100Hz is wasteful:** PX4's internal rate controller runs at 250Hz, but the offboard position loop doesn't need to match it. 100Hz over MicroXRCE-DDS increases UART bandwidth and CPU load on the companion computer without meaningful tracking improvement.
- **50Hz is optimal:** 20ms cycle time, 6cm spatial resolution at 3 m/s. Matches PX4 offboard trajectory mode specifications. Verified: `test_px4_offboard_controller.py` tests confirm 50Hz setpoint publishing.

Tone: Quick and decisive. This is a "know your numbers" question. Answer in under 30 seconds.

Q11: "If I kill the mesh node right now, what breaks? Which drone fails first?"

Testing: Failure mode awareness, graceful degradation understanding

GROUNDING ANSWER

Nothing fails immediately. Here's the degradation timeline:

- **t=0:** Mesh node dies. No new odometry broadcasts.
- **t=0 to 1.5s:** ORCA continues using last-known peer positions (within time horizon τ). All drones fly normally.
- **t=1.5s to 5s:** ORCA marks stale peers as "unknown." Drones switch to conservative velocity clamping (max 1.5 m/s). Search pattern degrades but continues.
- **t > 5s:** SwarmRAFT detects missing heartbeats. Remaining 4 drones repartition search sectors. The 5th drone (if isolated) enters autonomous RTL.
- **GCS:** Loses telemetry feed for the isolated drone. The operator sees "CONNECTION LOST" alert.

No drone "fails first" — they all degrade simultaneously because ORCA is decentralized. The mesh node is only needed for SwarmRAFT consensus and GCS telemetry, not for collision avoidance.

Tone: Calm and systematic. Walk through the timeline. The jury wants to see you've thought about failure modes.

Q12: "What's the one test you're MOST confident in, and one you're LEAST confident in?"

Testing: Self-awareness, honesty, depth of testing knowledge

GROUNDING ANSWER

Most confident: `test_orca_avoidance.py` — the 5-drone ring crossing clearance test. It verifies that 5 drones crossing simultaneously maintain $\geq 3.5\text{m}$ clearance. This is the core safety claim and it passes deterministically every time. The math is simple enough to verify by hand.

Least confident: `test_subsystem_b_full_integration.py` — this test imports from both Subsystem A and Subsystem C. If either package isn't colcon-built, the test fails with an import error, not a logic error. It tests integration wiring, not the algorithm itself. We should mock those imports but haven't yet.

Tone: Honest and self-aware. The jury RESPECTS "I know my weakest test" more than "all 232 tests are perfect."

III. Phase 2: Cross-Subsystem Integration

Questions 13–20 | Tests: Data flow across $A \leftrightarrow B \leftrightarrow C \leftrightarrow D$ | Difficulty: Hard

Q13: "Walk me the data path from camera pixel to GCS screen. Every hop."

Testing: Full-stack understanding, bandwidth awareness

GROUNDING ANSWER

1. **Camera sensor** publishes `sensor_msgs/Image` at 30Hz on `/camera/image_raw`.
2. **Subsystem C** (`detector_node.py`) subscribes, runs YOLOv8 detection, produces `/sutra/perception/targets` (JSON String at 10Hz).
3. **Subsystem B** (`mesh_node.py`) subscribes to targets, wraps in binary mesh protocol (`binary_mesh_protocol.py`), transmits over simulated 802.11s.
4. **Subsystem B** (`gcs_gateway_bridge.py`) receives targets, publishes to WebSocket on port 9090.
5. **Subsystem D** (React frontend) receives WebSocket message, updates survivor alert stream and 3D map markers.

Bandwidth at each stage:

Stage	Message Size	Rate	Bandwidth
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Camera Image	~600 KB	30 Hz	18 MB/s
YOLO Detections	~2 KB	10 Hz	20 KB/s
Binary Mesh	~64 bytes	10 Hz	640 B/s
WebSocket JSON	~500 bytes	10 Hz	5 KB/s

Bottleneck: Camera image to YOLO — 18 MB/s on the edge device. That's why we run YOLO locally and only transmit detection results, not raw video. The Deep JSCC path transmits compressed video separately for the operator view.

Tone: Fast and visual. Draw the pipeline on the whiteboard as you explain. The jury remembers diagrams.

Q14: "Your WebSocket bridge — is it JSON or binary? What's the message size at 50Hz?"

Testing: Engineering tradeoff awareness, serialization cost understanding

GROUNDING ANSWER

Both. The `gcs_gateway_bridge.py` supports two modes:

- **JSON mode** (default): Telemetry as JSON strings. ~500 bytes per message at 10Hz = 5 KB/s per drone. Easy to debug, human-readable.
- **Binary mode** (activated on high load): Uses `binary_mesh_protocol.py` with struct-packed payloads and CRC-32. ~64 bytes per message = 640 B/s per drone.

We use JSON for the GCS dashboard (SubsystD reads JSON natively) and binary for the mesh link (Subsystem B). The bridge translates between them.

At 50Hz: We don't send telemetry at 50Hz over WebSocket — we throttle to 10Hz using `telemetryBuffer.ts` (ring buffer). The 50Hz is for internal ROS 2 node-to-node communication, not GCS display.

Tone: Practical and specific. Show you understand the difference between internal (ROS 2) and external (WebSocket) data rates.

Q15: "What happens during RF mesh network partitioning? Split-brain?"

Testing: Distributed systems knowledge, consensus understanding

GROUNDING ANSWER

SwarmRAFT requires strict majority quorum: $Q = \lfloor N/2 \rfloor + 1 = 3$ out of 5 drones.

- **3-2 split:** The 3-drone partition continues as the active swarm. The 2 isolated drones enter autonomous RTL or climb to line-of-sight altitude to re-establish mesh.
- **2-2-1 split:** No partition has quorum. All drones enter RTL. This is the safe default — better to abort than operate without consensus.
- **3-1-1 split:** The 3-drone partition has quorum and continues. The 2 isolated drones RTL independently.

Key design decision: SwarmRAFT only governs task allocation and leader election, NOT collision avoidance. ORCA works independently. So even in split-brain, drones don't collide — they just can't coordinate search patterns.

Tone: Systematic. Enumerate the partitions. Show you've thought about every topology.

Q16: "How does the GCS handle a 6th drone? Does it still work?"

Testing: Scalability awareness, resource limits

GROUNDING ANSWER

Honest answer: We've tested with 5 drones. The GCS dashboard has 5 hardcoded drone tabs in `App.tsx`. Adding a 6th drone would require:

- Adding a tab in the React component (trivial).
- The WebSocket bridge already dynamically subscribes to `/ {drone_id} /` topics — it would auto-discover a 6th drone.
- The Canvas 2D rendering would need to handle 6 video streams — we haven't load-tested beyond 5.

What we know: The ORCA solver tested up to 50 drones in `test_huge_swarm_coordination.py`. The bottleneck is the GCS frontend, not the backend.

Tone: Honest about the limit. "We've validated 5. We believe it scales to more but haven't benchmarked it."

Q17: "Your emergency RTL — is it tested? What's the actual latency?"

Testing: Safety-critical feature verification

GROUNDING ANSWER

The RTL path is tested in multiple test files:

- `test_back_to_base_rtl.py` — 8 tests covering RTL dispatch, precision landing, motor failure fallback.
- `test_gcs_gateway_bridge.py` — Tests WebSocket RTL command dispatch.

Measured latency: RTL command from GCS click to PX4 VehicleCommand publish: ~3.8ms (measured in `test_subsystem_b_full_integration.py`).

Known gap: The GCS frontend sends RTL via WebSocket but has no confirmation loop — the operator doesn't get feedback if the command was received. This is a known issue we're fixing before Eval 3.

Tone: Show the test file. "Let me run it live." Live demonstrations are worth 10x more than verbal claims.

Q18: "How does SUTRA handle GPS-denied environments? Show me the code."

Testing: Core problem alignment, VIO implementation depth

GROUNDING ANSWER

GPS denial is handled at two levels:

1. **PX4 EKF2 level:** When `vehicle_gps_position.fix_type < 3`, PX4 automatically switches fusion sources via `EKF2_AID_MASK`. VIO pose and velocity are injected into `/fmu/in/vehicle_visual_odometry` at 50Hz.
2. **SUTRA VIO node:** `vio_localization.py` implements EKF2 VIO/GPS/Baro/Mag fusion with GPS-denied failover. When GPS drops, it switches to VIO-only mode with dead reckoning.

Code reference: `vio_localization.py` — the `VIOLocalizationNode` class. Test coverage: `test_vio_failover.py` — 5 tests covering EKF2 initialization, GPS primary mode, GPS drop → VIO failover, dead reckoning.

Limitation: VIO accumulates drift over time (~1.5% of distance traveled). In a real Kedarnath scenario with 10-minute missions at 3m/s, that's ~27m drift. We mitigate this with periodic GPS re-acquisition when available.

Tone: Point to the exact file. Open it on screen. "Here's the failover logic, line 200." The jury wants to see you can navigate your own codebase.

Q19: "If I add RF noise to the mesh, what degrades first?"

Testing: RF resilience understanding, failure priority awareness

GROUNDING ANSWER

Order of degradation:

1. **First (0-5% packet loss):** Deep JSCC video stream degrades gracefully (that's the whole point — analog-like degradation).
2. **Second (5-15% packet loss):** SwarmRAFT heartbeats start missing. Leader election timeout triggers. Non-critical (non-leader) nodes may briefly become candidates.
3. **Third (15-30% packet loss):** ORCA peer odometry goes stale. Drones switch to conservative velocity mode. Search pattern degrades.
4. **Fourth (>30% packet loss):** SwarmRAFT quorum may be lost. All drones enter RTL.

The key insight: Our system is designed so that *the most safety-critical function (collision avoidance) is the last to degrade*. ORCA only needs 10Hz peer updates, and it can operate on stale data for up to 1.5s. Deep JSCC video degrades first — which is acceptable because the operator can still see degraded imagery.

Tone: Layered and systematic. Show you've thought about the degradation cascade.

Q20: "How do the subsystems talk to each other? Is it all ROS 2 topics?"

Testing: IPC mechanism understanding, middleware awareness

GROUNDING ANSWER

Three communication mechanisms:

1. **ROS 2 Topics (intra-machine):** All nodes on the same companion computer communicate via DDS topics. This is the primary channel for GNC ↔ Perception ↔ Comms.
2. **WebSocket (inter-machine):** The `gcs_gateway_bridge.py` node bridges ROS 2 topics to WebSocket for the GCS frontend. This crosses the machine boundary (companion computer → operator laptop).
3. **Binary Mesh Protocol (inter-drone):** The `binary_mesh_protocol.py` handles struct-packed messages with CRC-32 for drone-to-drone communication over the Wi-Fi mesh.

Key design decision: We use ROS 2 DDS for reliability within a machine, WebSocket for cross-machine browser compatibility, and raw binary for bandwidth-constrained mesh links. Each mechanism is chosen for its strength.

Tone: Clear and architectural. Draw the 3-layer diagram on whiteboard.

IV. Phase 3: Problem Alignment & Real-World Validity

Questions 21–28 | Tests: 20-mark Problem Alignment criterion | Difficulty: Medium

Q21: "This runs in Gazebo with perfect GPS. How do I trust this works in a real Kedarnath disaster?"

Testing: Sim-to-real awareness, honesty about limitations

GROUNDNED ANSWER

You're right — Gazebo is a simulation. Here's what we've done to bridge the gap:

- **VIO failover tested:** `test_vio_failover.py` simulates GPS denial and verifies the drone switches to VIO without crashing.
- **Wind shear tested:** `test_wind_response.py` injects 12 m/s wind gusts and verifies position recovery.
- **ORCA tested with 50 drones:** `test_huge_swarm_coordination.py` verifies collision avoidance at scale.

What we CAN'T test in simulation: Real RF multipath in mountain terrain, actual thermal sensor noise, real-world GPS multipath under canopy. These require field testing.

Honest assessment: We're 80% of the way to a deployable system. The remaining 20% is hardware integration and field validation. That's what the hackathon 48-hour sprint is for.

Tone: Respectful of the question. "That's a fair concern." Then show what you HAVE validated.

Q22: "Name one thing about your system that would FAIL in a real deployment."

Testing: Intellectual honesty, self-awareness

GROUNDING ANSWER

Our Deep JSCC pipeline hasn't been tested on real SDR hardware. The neural network is trained and the inference runs on GPU, but we haven't connected it to a physical radio. In a real deployment, we'd need to:

- Map the 16 latent symbols to actual OFDM subcarriers.
- Handle carrier frequency offset and timing synchronization.
- Deal with real-world RF impairments (PA nonlinearity, phase noise).

This is the biggest gap between our simulation and a deployable system. For the hackathon, we're demonstrating the neural compression and PSNR curves — the SDR integration is a post-hackathon milestone.

Trap: If you say "nothing will fail," the jury loses all trust. Always have a real, specific answer ready.

Tone: Confident honesty. "I know exactly where the gap is, and here's my plan to address it."

Q23: "You mention NDMA. Which specific guideline governs UAV search patterns in flood zones?"

Testing: CONOPS depth, research thoroughness

GROUNDING ANSWER

Honest answer: NDMA doesn't have a specific guideline number for UAV search patterns — their guidelines focus on overall disaster response frameworks. Our CONOPS ([docs/conops/CONOPS_NDMA_Rescue_Profiles.md](#)) maps to:

- **NDMA Guidelines for Management of Earthquakes:** Search corridor patterns we adapted for aerial SAR.
- **NDMA Flood Response Framework:** Our Kedarnath flood profile follows the rapid assessment phase structure.
- **ICAO SAR Manual (Doc 8291):** Our search pattern spacing (100m track spacing at 30m AGL) follows ICAO conventions.**

We didn't invent the search patterns — we adapted established SAR doctrine for autonomous drone execution.

Tone: "We mapped our approach to existing standards" is better than "we invented a new standard." Show research depth without claiming false authority.

Q24: "If I give you 10 minutes to add one feature you don't have, what would it be?"

Testing: Priority awareness, architectural understanding

GROUNDNED ANSWER

Real batman-adv mesh networking. Right now, our 802.11s mesh is simulated in Python — we calculate FSPL, SNR, and packet loss mathematically, but there are no actual packets flowing over real radio hardware. Integrating batman-adv over mac80211_hwsim would:

- Validate that our FSPL model matches real RF behavior.
- Test actual packet delivery ratios under node churn.
- Prove the mesh works in a real multi-drone scenario.

The scripts exist (`scripts/setup_batman_mesh.sh` , `scripts/setup_mac80211_hwsim_mesh.sh`) but we haven't run them on real hardware yet.

Tone: Specific and actionable. Not "I'd add more features" but "I'd add THIS specific thing for THIS specific reason."

Q25: "How many people on your team wrote code vs. just wrote docs?"

Testing: Team contribution honesty, credit distribution

GROUNDING ANSWER

All 5 team members contribute to the codebase, but with different focuses:

- Nikhil: Subsystem A (GNC — 21 Python modules, 120 tests) and Subsystem B (Comms — 3 ROS 2 nodes, 48 tests). Primary architect.
- Vedanth: Subsystem C (Perception — 2 ROS 2 nodes, 5 modules, 60 tests). WGS84 raycaster, ByteTrack, YOLO integration.
- Siva: Subsystem D (GCS — React 18 frontend, 10 source files, ~3200 lines). WebSocket integration, Canvas physics sims.
- Harika: Subsystem E — test verification scripts, DOCS.md maintenance, pitch deck, presentation design. Also contributed to the verification agent (`scripts/sutra_verification_agent.py`).
- Rohith: Subsystem F — CONOPS, field deployment SOPs, NDMA rescue profiles. Also runs GPU compute workloads for Vedanth.

Harika and Rohith's contributions are documentation and operational planning — which is exactly what Subsystems E and F are assigned to in our architecture.

Tone: Give credit generously. Never say "only 3 of us coded." Every role matters.

Q26: "What dataset did you train YOLO on? How many images? Is it your own data?"

Testing: ML training depth, data provenance

GROUNDING ANSWER

We fine-tuned YOLOv8-Nano on two public datasets:

- VisDrone: 10,209 images of aerial pedestrian/vehicle detection. Public benchmark dataset from Tianjin University.
- HIT-UAV: 2,886 thermal images of person detection. Public dataset from Harbin Institute of Technology.

Training details: `scripts/train_visdrone_yolo.py` — trained on Kaggle GPU (T4). 50 epochs, batch size 16, imgsz 640. Output: `best.pt` (5.92 MB), mAP@0.5 = 94.2%.

We did NOT collect our own data. We used public benchmarks and fine-tuned. For a real deployment, we'd need to collect SAR-specific data from actual disaster scenarios — that's a post-hackathon effort.

Tone: Honest about data provenance. "We used public benchmarks" is perfectly valid for a hackathon.

Q27: "What's the total compute power needed to run your system on one drone?"

Testing: Hardware feasibility, SWaP-C awareness

GROUNDING ANSWER

Component	CPU Usage	RAM	GPU
PX4 Autopilot (FMU)	~5% (ARM Cortex)	2 MB	None
ROS 2 + GNC Nodes	~15%	200 MB	None
YOLOv8 TensorRT	~5%	300 MB	2 GB VRAM
ByteTrack MOT	~3%	50 MB	None
Deep JSCC Encoder	~2%	100 MB	1 GB VRAM
SwarmRAFT + Mesh	~5%	80 MB	None
Total	~35%	~730 MB	~3 GB VRAM

Minimum hardware: Jetson Orin Nano (40 TOPS, 8GB RAM) or Orange Pi 5 + RTX 3050. Both are under ₹15,000.

Tone: Quick table. The jury wants to know "can this fit on a drone?" — answer with numbers, not hand-waves.

Q28: "Why should I use your system instead of just using one good drone with a thermal camera?"

Testing: Value proposition clarity, swarm justification

GROUNDING ANSWER

Three reasons:

- Coverage:** One drone with a 70° FOV camera at 30m AGL covers a 35m wide corridor. Kedarnath flood ravines are 200-500m wide. One drone takes 10+ passes. Five drones cover it in 2 passes — 5× faster search in the golden hour after a disaster.
- Fault tolerance:** If one drone crashes (motor failure, RF jamming, battery), the mission continues with 4 drones. A single drone is a single point of failure.

3. Cross-validation: Multiple drones see the same target from different angles. Our fusion engine uses this for 3D localization — a single drone can only give 2D bearing, not 3D position.

The math: Search area coverage rate = drone_count × (speed × FOV_width). 5 drones at 3m/s × 35m FOV = 525 m²/s. One drone = 105 m²/s. That's the difference between finding someone in 10 minutes vs. 50 minutes.

Tone: Lead with the disaster scenario. "In the golden hour after a landslide, 40 minutes matters." Make it visceral.

V. Phase 4: Innovation & Mathematical Rigor

Questions 29–36 | Tests: 15-mark Innovation criterion | Difficulty: Hard

Q29: "ORCA or Artificial Potential Fields — you chose ORCA. If I give you a concave obstacle, does ORCA still work?"

Testing: Algorithm limitation awareness

GROUNDING ANSWER

No. ORCA doesn't handle concave obstacles well. Neither do Potential Fields. This is a known limitation of both approaches.

Our solution: The 3D Occupancy Grid (`sutra_fsd_occupancy.py`) pre-filters the environment. Before ORCA runs, the occupancy grid marks concave regions (caves, U-shaped ravines) as occupied. The trajectory planner routes around them. ORCA only handles dynamic drone-drone avoidance — static obstacle avoidance is done by the grid.

This is a deliberate architectural choice: separate static obstacle avoidance (grid) from dynamic avoidance (ORCA). Combining them into one algorithm would be more complex and harder to verify.

Tone: Don't be defensive. "You're right — and here's how we handle it." Honesty about limitations is a strength.

Q30: "Derive the ORCA half-plane formulation on the whiteboard. Right now."

Testing: Math under pressure, derivation ability

GROUNDING ANSWER — WHITEBOARD DERIVATION

Write step by step:

1. Velocity Obstacle: $VO_{ij} = \{v \mid \exists t \in [0, \tau], t \cdot v \in B(p_j - p_i, r_i + r_j)\}$
"The set of velocities that will cause drone i to collide with drone j within time τ ."
2. Minimum correction: $u = \operatorname{argmin}_{w \in \partial VO} \|w - (v_i - v_j)\| - (v_i - v_j)$
"The smallest velocity change to exit the velocity obstacle."
3. ORCA half-plane: $ORCA_{ij} = \{v \mid (v - (v_i + \frac{1}{2}u)) \cdot n \geq 0\}$
"The permissible velocities for drone i, sharing the correction equally between both drones."

The $\frac{1}{2}$ factor is the "reciprocal" in ORCA — both drones share the collision avoidance responsibility equally.

Tone: Write slowly. Explain each line. Make eye contact after each step. This is a performance — the jury wants to see you can DERIVE, not just recite.

Q31: "You have 232 tests. What percentage are unit tests vs. integration tests?"

Testing: Testing methodology awareness

GROUNDING ANSWER

Approximate breakdown:

- Unit tests (~80%): Test individual algorithms in isolation. Examples: ORCA clearance math, quintic spline continuity, GPS raycast formula, ByteTrack IoU computation, CBF barrier enforcement. No ROS 2 nodes spun up.
- Integration tests (~15%): Test ROS 2 node instantiation, topic wiring, parameter handling. Examples: `test_subsystem_b_full_integration.py`, `test_gcs_gateway_bridge.py`.
- Stress tests (~5%): Test extreme conditions. Examples: 100-node mesh, 50-drone ORCA, 1000-frame ByteTrack, 10000-struct binary protocol alignment.

We follow the testing pyramid — many fast unit tests at the base, fewer integration tests, and targeted stress tests at the top.

Tone: Quick and structured. Show you understand testing methodology, not just test count.

Q32: "What's the battery life of one drone running your full stack?"

Testing: Practical deployment awareness

GROUNDING ANSWER

Honest answer: We haven't measured actual flight time with the full stack running.

What we know:

- A standard 5" quadcopter with 4S 1500mAh battery gets ~15 minutes hover time without compute payload.
- Adding Jetson Orin Nano (~15W) reduces that to ~10-12 minutes.
- Our compute stack draws ~35% CPU + 3GB VRAM — that's roughly 10-15W additional.
- Estimated flight time with full stack: 8-10 minutes.

This is why we need 5 drones — not just for coverage, but for relay. While 3 search, 2 recharge. Our CONOPS includes a rotation schedule.

Tone: Honest about what you haven't measured, then provide the engineering estimate. Show you've thought about it even without a physical test.

Q33: "What's the maximum range of your mesh network between drones?"

Testing: RF range awareness, real-world deployment constraints

GROUNDING ANSWER

Our FSPL model (`mesh_node.py`) calculates:

$$\text{FSPL(dB)} = 20 \cdot \log_{10}(d_{\text{km}}) + 20 \cdot \log_{10}(f_{\text{MHz}}) + 32.44$$

At 2.4 GHz Wi-Fi with 20 dBm TX power:

- 50m range: FSPL = 60 dB, SNR = 40 dB, PDR ≈ 100%
- 100m range: FSPL = 66 dB, SNR = 34 dB, PDR ≈ 99%
- 200m range: FSPL = 72 dB, SNR = 28 dB, PDR ≈ 95%
- 500m range: FSPL = 80 dB, SNR = 20 dB, PDR ≈ 80%

Practical range: 200-300m with reliable mesh. For the Kedarnath scenario with 200m-wide ravines, 5 drones at 50m spacing stay well within range.

Caveat: This is FSPL (free-space path loss). Real mountain terrain adds 10-20 dB of shadow fading. Our conservative estimate is 150m reliable range in mountainous terrain.

Tone: Show the math, then caveat with real-world factors. The jury respects "our model says X but reality adds Y."

Q34: "What's the false positive rate of your survivor detection?"

Testing: Perception accuracy awareness, SAR operational impact

GROUNDING ANSWER

We don't have a measured false positive rate. Here's why and what we know:

- $mAP@0.5 = 94.2\%$ on VisDrone test set. This means 94.2% of ground-truth objects are detected with $IoU \geq 0.5$.
- Precision (how many detections are correct) is approximately 87% on the VisDrone validation set — meaning ~13% of detections are false positives.
- In SAR context: A 13% false positive rate means the operator gets ~1 false alarm per 8 real detections. That's acceptable for a first-pass search — the operator visually confirms each alert.

We deliberately bias toward high recall (catching all survivors) over high precision (fewer false alarms). Missing a survivor is catastrophic; a false alarm wastes 30 seconds of operator time.

Tone: Honest about the tradeoff. "We chose recall over precision because in SAR, missing a person is worse than a false alarm."

Q35: "What's the latency from camera capture to detection on screen?"

Testing: End-to-end latency awareness

GROUNDING ANSWER

Stage	Latency
Camera exposure + readout	~33ms (30Hz)
USB/CSI transfer to CPU	~5ms
YOLOv8 TensorRT inference	~5ms
ByteTrack association	~1ms
ROS 2 topic publish	~1ms
WebSocket serialization	~2ms

Browser rendering	~16ms (60 FPS)
Total	~63ms

At 3 m/s flight speed, 63ms = 19cm of flight distance. The detection appears on screen where the drone was 19cm ago. For SAR at 30m AGL, this is acceptable.

Tone: Quick table. Show you've measured (or at least estimated) every stage.

Q36: "What's the most novel thing about your system that no other team at this hackathon will have?"

Testing: Competitive awareness, moat articulation

GROUNDING ANSWER

Deep JSCC for drone swarm communications. Most drone projects use standard Wi-Fi or H.264 video streaming. In a real disaster with RF jamming or severe multipath, H.264 hits the digital cliff and freezes completely.

Our Deep JSCC autoencoder provides *graceful degradation* — as signal quality drops, the video gets progressively blurrier but never freezes. The operator always sees something, even at -5 dB SNR where H.264 shows nothing.

This is not just a clever hack — it's a fundamentally different approach to wireless video that's backed by a trained neural network, real PSNR measurements, and a published research pipeline.

Tone: Confident but not arrogant. "We believe this is unique" is better than "nobody else can do this."

VI. Phase 5: Team Dynamics & Role Probing

Questions 37–42 | Tests: Team depth, communication, credit distribution

Q37 (to Harika): "You're the docs lead. Explain to me like I'm 5 — what does your drone actually DO when it finds a person?"

Testing: Non-coder articulation, cross-functional depth

GROUNDING ANSWER (HARIKA)

"Imagine you're looking for someone lost in a flood. Our drone flies over the water and takes pictures with two cameras — one regular camera and one heat camera that sees body warmth even through smoke or darkness.

When the drone spots what looks like a person, it sends a message to the other 4 drones: 'I found something here!' All 5 drones then fly to that area to confirm — like friends helping search together.

At the same time, the drone calculates exactly where on the ground that person is — not just 'somewhere below' but exact GPS coordinates accurate to less than a foot. This location appears on the rescue team's map screen within seconds.

The rescue team never has to manually pilot the drones. The drones decide together where to search, find people, and report back — all on their own."

Tone (Harika): Warm, clear, no jargon. Use hand gestures. The jury wants to see that EVERY team member understands the system, not just the coders.

Q38 (to Siva): "Your GCS shows a map. Why should I use your map instead of Google Maps?"

Testing: GCS value proposition, differentiation awareness

GROUNDING ANSWER (SIVA)

"Three reasons:

1. No internet needed: Google Maps requires internet. In a disaster zone, cell towers are down. Our GCS runs on localhost — completely offline.
2. Real-time drone positions: Google Maps doesn't show 5 drones flying in formation with live battery levels, mesh health, and search progress. Ours does.
3. Survivor alerts: When a drone detects a person, the marker appears on our map instantly with a thermal image thumbnail and GPS coordinates. Google Maps can't do that."

Think of it like this: Google Maps is a road map. Our GCS is a tactical operations dashboard — purpose-built for disaster response."

Tone (Siva): Confident, practical, operator-focused. "What would a rescue worker need?" is the right framing.

Q39 (to Vedanth): "Your YOLO model — if I show it a picture of a dog, does it think it's a survivor?"

Testing: Model specificity understanding, training data awareness

GROUNDING ANSWER (VEDANTH)

"No — our model is trained specifically on aerial person detection. It won't detect a dog at all. The VisDrone dataset only contains person, car, van, bus, truck, and bicycle classes.

But here's the important nuance: in our fusion engine, we have a class filter — `SAR_CLASS_IDS` — that only passes person-class detections through. So even if YOLO somehow detected a dog, it would be filtered out before reaching the operator.

For a real deployment, we'd want to retrain on SAR-specific data — people in rubble, under canopies, partially submerged. The VisDrone dataset is a starting point, not the final answer."

Tone (Vedanth): Technical but accessible. Show you understand the model's limitations as well as its capabilities.

Q40 (to Rohith): "You're the ops lead. What happens if it rains during the demo?"

Testing: Operational planning, contingency thinking

GROUNDING ANSWER (ROHITH)

"For the hackathon demo, everything runs on laptops — Gazebo Sim 8, the GCS dashboard, the test suite. Rain doesn't affect software.

For a real deployment, our CONOPS includes weather minimums:

- Wind: max 15 m/s (our SutraNeuroFlight handles 18 m/s, so we have 3 m/s margin).
- Rain: Light rain is fine — our thermal camera sees through it. Heavy rain degrades optical but thermal still works.
- Visibility: We don't need visual visibility — the autonomous stack uses VIO and radar.

The field deployment SOP (`docs/conops/Field_Deployment_SOP.md`) has a pre-flight weather check in Stage B."

Tone (Rohith): Practical, operator-focused. "Here's what we do" not "here's what we'd hope for."

Q41: "Who reviewed the code? Did anyone do code review?"

Testing: Engineering maturity, collaboration awareness

GROUNDING ANSWER

"We have a 3-tier branch structure: feature branches → dev (sandbox) → main (verified). Every merge to main requires:

1. All 232 tests passing.
2. At least one other team member reviewing the diff.
3. DOCS.md updated with new benchmarks.

For the hackathon, we're following the same discipline. Nikhil has cross-branch access as Tech Lead for emergency fixes, but all other merges go through the review flow."

Tone: Show engineering maturity. "We take code quality seriously even under time pressure."

Q42: "What did each person build? Point to specific files."

Testing: Individual contribution verification

GROUNDING ANSWER

Member	Key Files
Nikhil	<code>orca_avoidance.py</code> , <code>px4_offboard_controller.py</code> , <code>sutra_fsd_autopilot_node.py</code> , <code>mesh_r</code> <code>perceptron_jscc.py</code>
Vedanth	<code>detector_node.py</code> , <code>bytetrack.py</code> , <code>camera_streamer_node.py</code>
Siva	<code>App.tsx</code> , <code>GisTelemetryHud.tsx</code> , <code>SwarmRingCrossingArena.tsx</code>
Harika	<code>scripts/sutra_verification_agent.py</code> , <code>docs/hackathon/JURY_FEEDBACK_TRACKER.md</code>
Rohith	<code>docs/conops/CONOPS_NDMA_Rescue_Profiles.md</code> , <code>docs/conops/Field_Deployment_SOP.md</code>

Tone: Quick, factual, point to the files on screen. Let the code speak.

VII. Phase 6: The Breaker Questions

Questions 43–50 | Highest stress | Tests composure under attack

Q43: "I see a hardcoded GPS origin for San Francisco in your code. We're in Bengaluru. Is this project real?"

Testing: Honesty about known gaps, composure under attack

GROUNDING ANSWER

"You're right — `detector_node.py:130` has `ORIGIN_LAT=37.774929`. That's the default value for Gazebo SITL simulation. In our launch files, we override it with site-specific coordinates via ROS parameters.

For the Bengaluru demo, we'd set it to the NHCE campus coordinates. Let me show you the parameter override..."

(If you can't show the override: "We haven't added the launch file parameter override yet — it's on our task list before Eval 3. The parameter exists in the code, we just need to set it in the launch file.")

Tone: Calm. Don't get defensive. "You're right, here's how we handle it" — then show the code.

Critical: If you can't show the parameter override, admit it. "We'll fix this before Eval 3" is better than pretending.

Q44: "Your package.xml is missing numpy, torch, and websockets as dependencies. Did you actually run this?"

Testing: Dependency management awareness

GROUNDING ANSWER

"The ROS 2 dependencies are declared correctly — `rcldpy`, `sensor_msgs`, `std_msgs`, `geometry_msgs`. The Python ML dependencies (numpy, torch, websockets) are installed in our Docker environment and `requirements.txt`, but not declared in `package.xml` as `exec_depend`.

That's a packaging gap we should fix. For the hackathon, our Docker image has everything pre-installed. But you're right — for a clean `colcon` build on a fresh machine, those would fail."

Tone: Acknowledge and explain. "That's a valid catch — here's our workaround."

Q45: "I'm going to pick a random file and explain every line. Pick a file."

Testing: Deep code knowledge, composure under randomness

GROUNDING ANSWER

Pick `orca_avoidance.py` — you know it cold.

"This file implements the 3D ORCA collision avoidance solver. Let me walk through it:

- Lines 1-50: Imports and data classes. `AgentState` holds position, velocity, and radius for each drone.
- Lines 51-150: `Orca3DSolver.compute_avoidance()` — the core algorithm. For each peer drone, it computes the velocity obstacle cone, finds the closest point on the cone boundary, and constructs the ORCA half-plane constraint.
- Lines 151-250: Linear programming solver — finds the velocity closest to the preferred velocity that satisfies all half-plane constraints simultaneously.
- Lines 251-350: SORCA extension — adds acceleration bounding for smooth motor commands.
- Lines 351-500: ROS 2 node — subscribes to peer odometry, runs the solver at 10Hz, publishes avoidance velocities."

Tone: Confident and methodical. You BUILT this — own it. Navigate to the file and scroll through as you explain.

Q46: "What's the one thing you'd do differently if you started over?"

Testing: Reflection, learning ability, architectural awareness

GROUNDING ANSWER

"I'd integrate the real mesh networking earlier. We spent a lot of time building a beautiful Python simulation of 802.11s — FSPL models, SNR calculations, packet loss estimation — but it's all analytical. If I started over, I'd set up `batman-adv` over `mac80211_hwsim` in week 1 and validate our models against real packet delivery from day one.

The lesson: simulation is necessary but not sufficient. You need to validate the model against reality as early as possible."

Tone: Reflective and genuine. Show you learn from experience.

Q47: "Your test count says 232 but I see 228 in one place and 232 in another. Which is it?"

Testing: Precision, consistency, number integrity

GROUNDING ANSWER

"The correct count is 232. The 228 figure in the pitch deck was from an earlier run before we added the latest test files. Let me run it live to confirm..."

(Run: `pytest sutra_ws/src/sutra_*/test/ --co -q 2>&1 | tail -1`)

"232 tests collected. The discrepancy was a documentation sync issue — we updated the code but not the slide."

Tone: Don't argue. Show the live count. "Let me verify" is always stronger than "trust me."

Q48: "If I press you — is any of this code copied from open source?"

Testing: Originality verification, NHCE Rule 6.2 compliance

GROUNDING ANSWER

"The algorithms are inspired by published papers — ORCA from van den Berg et al. 2011, ByteTrack from Zhang et al. 2022, CBF from Ames et al. 2014. But the implementation is original.

For example, our ORCA solver (`orca_avoidance.py`) implements 3D reciprocal velocity obstacles with SORCA acceleration bounding — that specific combination doesn't exist in any open-source library. We wrote it from scratch based on the paper's mathematical formulation.

Our Deep JSCC autoencoder architecture is original — the Swin Window Attention integration with channel-blind JSCC is our own design, not from any existing repository.

We cite all third-party libraries in our README per NHCE Rule 6.4.1."

Tone: Confident but specific. "Inspired by papers, implemented from scratch" is the right framing.

Q49: "What happens if two drones detect the same survivor simultaneously?"

Testing: Multi-drone coordination awareness, deduplication logic

GROUNDING ANSWER

"Our fusion engine handles this at two levels:

1. Local deduplication: ByteTrack MOT assigns a unique track ID to each detected person. If two drones see the same person, they independently assign different track IDs — but the GPS coordinates will be nearly identical (within 0.32m).
2. Swarm-level deduplication: The `mesh_node.py` aggregates targets from all drones. Targets with GPS coordinates within 2m of each other are merged into a single confirmed target with higher confidence."

The mesh deduplication logic is in `test_subsystem_b_c_wiring.py` — it tests that duplicate targets from multiple drones are correctly merged."

Tone: Show you've thought about the multi-drone edge case. "Great question — we handle this at two levels."

Q50: "We're done. Any questions for us?"

Testing: Confidence, engagement, strategic thinking

GROUNDING ANSWER

Never say "No." Always have a question ready. Options:

- "What's the one capability you'd want to see in a SAR drone system that most teams don't demonstrate?"
- "In your experience, what separates a hackathon project from a deployable system?"
- "Would you like to see the test suite run live, or would you prefer to dive deeper into a specific subsystem?"

The last option is strategic — it gives the jury a choice and shows you're confident enough to demo anything.

Tone: Grateful and engaged. "Thank you for the questions — we learned a lot from this conversation."

VIII. Psychological Patterns — How Juries Think

Recognize and respond to common jury pressure tactics

Jury Tactic	What They're Testing	How to Respond	What NOT to Do
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Silence after your answer	Confidence — do you fill the silence with rambling?	Stop talking. Let your answer stand. Take a sip of water if needed.	Don't say "uh... and also..." or re-explain.
"I don't understand"	Communication clarity — can you explain differently?	Rephrase with an analogy, not more jargon. "Think of it like this..."	Don't repeat the same technical explanation louder.
"That's not impressive"	Resilience — do you get defensive or double down with evidence?	"Let me show you the test output" — point to data, not opinions.	Don't argue or say "actually, it IS impressive because..."
Interrupting mid-sentence	Priority — can you get to the point faster?	Stop. Restart from the conclusion: "The short answer is X. The details are..."	Don't try to finish your sentence after being interrupted.
Asking the quiet team member	Team depth — is this a one-person show?	The quiet member MUST answer. Even if brief. "Vedanth, can you take this one?"	Don't have Nikhil jump in to answer for them.
"What's the weakest part?"	Honesty — do you pretend everything is perfect?	Name the real gap and your plan. "Our mesh is simulated — we're integrating real batman-adv during the hackathon."	Don't say "everything is strong" or pick a trivial weakness.
"Show me the code"	Implementation reality — do you actually know where things are?	Navigate to the exact file and line without hesitation.	Don't say "it's somewhere in the comms package."
Deliberately misstating your architecture	Will you agree with errors or correct them?	Correct politely: "Actually, it's slightly different — let me clarify."	Don't agree to be polite. Don't be confrontational either.
"This is just a demo, not real"	Defensiveness — can you acknowledge the gap without crumbling?	"You're right that simulation isn't deployment. Here's what we've validated in sim and what needs hardware testing."	Don't say "but our simulation is very accurate!"
	They're checking something you said		

Looking at their phone during your answer	against facts. Or they're just tired.	Keep talking. Your answer should be self-contained.	Don't stop and ask "are you listening?"
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IX. Tone & Delivery Masterclass

How to sound confident without being arrogant

The Three Tones

Tone	When to Use	Example
Teacher	Explaining architecture, walking through code	"Let me walk you through the pipeline..." (slow, methodical)
Engineer	Answering technical questions, citing numbers	"The ORCA solver runs in 0.5ms for 5 agents. Here's the benchmark..." (precise, data-driven)
Honest human	Admitting gaps, acknowledging limitations	"We haven't tested that yet. Here's what we know and what we don't..." (genuine, self-aware)

Voice and Body Language

- Speak at 70% of your normal speed. Under stress, you'll speed up. Consciously slow down.
- Pause after key points. "Our CBF safety shield maintains 2.80m clearance." [pause 1 second] Let it sink in.
- Make eye contact with the questioner while answering, then scan the rest of the jury.
- Stand straight. Hands visible. Don't cross arms, don't fidget, don't touch your face.
- When writing on whiteboard: Write the formula first, then turn around and explain. Don't talk while writing — the jury can't read and listen simultaneously.

Power Phrases

- "Let me show you" (instead of "let me explain")
- "The data shows" (instead of "I think")
- "Here's what we validated" (instead of "we believe")
- "You're right" (when they catch a gap — acknowledge, don't defend)
- "That's a great question" (only if it actually is — don't overuse)

Anti-Patterns

- "Um, basically, like..." — eliminates credibility. Replace with silence.
- "As I said earlier..." — sounds condescending. Rephrase instead.
- "Trust me" — never say this. Show, don't tell.
- "That's a good question" to every question — sounds robotic.
- Rushing through the answer — speed = nervousness. Slow down.

X. Do's and Don'ts — Quick Reference Card

DO	DON'T
Run pytest live and show terminal output	Read benchmark numbers from slides
Say "I don't know, but I'll find out" when you don't	Make up an answer
Point to the exact file:line when asked about code	Say "it's somewhere in the package"
Admit "our mesh is simulated, not real 802.11s"	Claim you have real mesh networking
Show the 5-UAV ring crossing in Gazebo	Show a pre-recorded video first
Let each member answer their pillar	Have Nikhil answer everything
Draw diagrams on the whiteboard	Point at the screen without drawing
Use specific numbers: "0.5ms, 3.80m, 50Hz"	Use vague terms: "very fast, very accurate"
Acknowledge gaps with a fix plan	Pretend gaps don't exist
Thank the jury after tough questions	Look annoyed or defensive

XI. Emergency Recovery — If You Don't Know the Answer

What to say when you're completely stuck

Scenario: You blank on a math derivation

Say: "The full derivation is in the code — let me show you the implementation instead of writing it on the board." Then open the relevant file and walk through the code logic.

Scenario: They ask about something you haven't built

Say: "That's not implemented yet. It's on our roadmap for the 48-hour sprint. Here's how we plan to approach it..." Then describe the architecture even if the code doesn't exist yet.

Scenario: They catch a real bug

Say: "You're right — that's a bug. Let me fix it right now." If you can fix it in 2 minutes, do it live. Nothing impresses a jury more than fixing a bug on stage.

Scenario: Two jury members disagree with each other

Say: "Both perspectives are valid. Our current implementation follows [approach A], which aligns with [jury member 1]'s point. We considered [approach B] but chose A because [reason]."

Scenario: The demo crashes

Say: "Let me restart the simulation." while calmly running the recovery command. Have the fallback terminal ready: `pytest sutra_ws/src/sutra_*/test/ -q` — tests are more impressive than a crashed demo.

Scenario: They ask a question outside everyone's expertise

Say: "That's outside my area of expertise. [Team member name] might be better positioned to answer." If nobody knows: "We'd need to research that. Can we follow up after the evaluation?"

Remember: The jury isn't trying to fail you.

They want to see that you understand your system deeply, can communicate clearly under pressure, and have the honesty to acknowledge what you don't know. That's what separates a real engineer from a slide deck.